Network Mini-project

Ray Tsai

We study the structural properties of the all-or-none network growth model, a variation of preferential attachment and copying mechanisms in which a new node connects to a uniformly chosen target and, with probability p, to *all* of its neighbours, or with probability $1-p$ to *none*. We derive analytical expressions for the expected number of edges as a function of p and the number of nodes N, characterise the degree distribution in the sparse regime, and compute the clustering coefficient, assortativity coefficient, and average shortest-path distance. Our analytical predictions are supported by extensive numerical simulations.

I. INTRODUCTION

Network growth models are central to our understanding of real-world complex systems, ranging from the World Wide Web to biological and social networks [1, 2]. The Barabási–Albert (BA) model [1] introduced preferential attachment as a mechanism for generating power-law degree distributions, while copying models [3] propose an alternative microscopic rule in which new nodes replicate the connections of existing ones.

In this work we analyze the *all-or-none* model: at each growth step a new node is added and connected to a target chosen uniformly at random; it then connects to *all* neighbours of the target with probability p, or to *none* of them with probability $1-p$. This binary choice creates a sharp interpolation between two qualitatively distinct regimes and leads to non-trivial structural properties that we characterise below.

The remainder of this paper is organised as follows. Section II defines the model precisely. Sections III–VII derive analytical results and present numerical evidence for the edge count, degree distribution, clustering, assortativity, and average shortest-path distance, respectively. Section VIII concludes.

II. MODEL DEFINITION

The network begins with a single node ($N = 1$). At each growth step ($N \to N+1$):

1. A new node v is introduced.
2. A target node u is chosen uniformly at random from the existing N nodes.
3. An edge (v, u) is added.
4. With probability p, edges (v, w) are added for every neighbour w of u; with probability $1-p$, no further edges are added.

III. NUMBER OF EDGES

Let M denote the expected number of edges in the network, and write M_N for the expected number of edges when the network has N nodes. At step $N \to N+1$, a target u is chosen uniformly from the N existing nodes. The expected number of new edges is $1 + p\,k_u$, where k_u is the degree of u. Note that by the handshake lemma, the average value of k_u over all choices of u is $2M_N/N$. It follows that,

$$\begin{aligned} M_{N+1} &= M_N + \left(\frac{2p}{N} M_N + 1\right) \\ &= \left(1 + \frac{2p}{N}\right) M_N + 1. \end{aligned} \tag{1}$$

This is a first-order linear recurrence. A particular solution of the form $\alpha_M N$ yields $\alpha_M = 1/(1-2p)$ for $p \neq 1/2$. The homogeneous solution is $\mathcal{A}_N = \Gamma(N+2p)/[\Gamma(1+2p)\,\Gamma(N)]$, satisfying $\mathcal{A}_{N+1}/\mathcal{A}_N = (N+2p)/N$. Imposing $M_1 = 0$ now yields

$$\boxed{M = \frac{1}{1-2p}\left(N - \frac{\Gamma(N+2p)}{\Gamma(1+2p)\,\Gamma(N)}\right)}, \tag{2}$$

for $p \neq 1/2$. Since $\Gamma(N+2p)/\Gamma(N) \sim N^{2p}$ as $N \to \infty$, two regimes emerge:

- *Sparse* ($p < 1/2$): $M \sim \frac{N}{1-2p}$ (linear).
- *Dense* ($p > 1/2$): $M \sim \frac{N^{2p}}{(2p-1)\,\Gamma(1+2p)}$ (superlinear).

A. Numerical results

FIG. 1. Number of edges M as a function of p for several values of N, obtained by averaging over 10^4 independent realisations. Solid lines show the analytical prediction (2).

IV. DEGREE DISTRIBUTION

Denote by $f(k, i, N)$ the probability that node i has degree k when the network has N nodes. At step $N \to N+1$, degree k increases by one with probability $(1+pk)/N$, because either the node is chosen as target (probability $1/N$) or a neighbour of it is chosen and the

"all" event occurs (probability pk/N) The master equation for f is then

$$f(k,i,N+1) = \frac{1+p(k-1)}{N} f(k-1,i,N) + \left(1 - \frac{1+pk}{N}\right) f(k,i,N). \quad (3)$$

Assume that $p < 1/2$ so that the expected number of edges $\mathbb{E}[M]$ is linear in N, by the results of Section III. This allows us to seek the asymptotic degree distribution $P_k = \lim_{N\to\infty} n_k(N)/N$, where $n_k(N) = \sum_{i=1}^{N} f(k,i,N)$ is the expected number of nodes with degree k in the N-node network. Note that the new node enters with degree 1 with probability $1-p$ and with degree k with probability $p\, n_{k-1}(N)/N$ for $k \geq 2$. Hence, summing (3) over the N existing nodes and adding the new node yields

$$n_k(N+1) = \frac{1+p(k-1)}{N} n_{k-1}(N) + \left(1 - \frac{1+pk}{N}\right) n_k(N) + (1-p)\,\delta_{k,1} + \frac{p\, n_{k-1}(N)}{N}. \quad (4)$$

Substituting $n_k(N) = NP_k$, we have

$$(2+pk)\, P_k = (1+pk)\, P_{k-1} + (1-p)\,\delta_{k,1}. \quad (5)$$

This yields $P_1 = (1-p)/(2+p)$ and, for $k \geq 2$,

$$P_k = \frac{1+pk}{2+pk} P_{k-1}, \quad (6)$$

which have the closed form

$$\boxed{P_k = \frac{1-p}{2+p} \prod_{j=2}^{k} \frac{1+pj}{2+pj}}, \quad k \geq 1. \quad (7)$$

In terms of the Gamma function,

$$P_k = \frac{(1-p)\,\Gamma(2+2/p)}{(2+p)\,\Gamma(2+1/p)} \cdot \frac{\Gamma(k+1+1/p)}{\Gamma(k+1+2/p)}. \quad (8)$$

As $k \to \infty$, we have $\Gamma(k+1+1/p)/\Gamma(k+1+2/p) \sim k^{-1/p}$, and so

$$P_k \sim \lambda_p k^{-1/p}, \quad (9)$$

where $\lambda_p = (1-p)\,\Gamma(2+2/p)/[(2+p)\,\Gamma(2+1/p)]$. The degree distribution thus follows a power law with exponent $1/p$, which matches the degree distribution of the BA model at $p = 1/3$.

V. TRIANGLES AND CLUSTERING

A. Triangles

Let T denote the expected number of triangles in the network, and write T_N for the expected number of triangles when the network has N nodes. Triangles are created in an all-or-none network if and only if the "all" event occurs and $N \geq 2$. If the target u has degree k_u and there are e_u edges among its neighbours, the new node v forms k_u triangles of type (v,u,w_i) and e_u triangles of type (v,w_i,w_j), where w_i, w_j are distinct neighbours of u. Note that $\sum_u k_u = 2M_N$ and $\sum_u e_u = 3T_N$. Averaging over the target choice u thus yields

$$T_{N+1} = \left(1 + \frac{3p}{N}\right) T_N + \frac{2p\, M_N}{N}. \quad (10)$$

This linear recurrence has the same structure as (1). Substituting the exact edge formula and solving with $T_1 = 0$ gives, for $p \neq 1/2, 1/3$,

$$\boxed{T = \frac{2pN}{(1-2p)(1-3p)} + \frac{2\mathcal{A}_N}{1-2p} - \frac{2\mathcal{B}_N}{1-3p}}, \quad (11)$$

where $\mathcal{A}_N$ is as in (2) and $\mathcal{B}_N = \Gamma(N+3p)/[\Gamma(1+3p)\Gamma(N)]$. Thus two regimes emerge as $N \to \infty$:

- $p < 1/3$: $T \sim \frac{2p}{(1-2p)(1-3p)} N$ (linear).
- $p > 1/3$: $T \sim \frac{2}{\Gamma(1+3p)(3p-1)} N^{3p}$ (superlinear).

B. Clustering coefficient

The global clustering coefficient is given as

$$C = \frac{3T}{W}, \quad (12)$$

where $W = \frac{1}{2}\sum_i k_i(k_i - 1)$ is the expected number of connected triples. Let $S_N = \sum_i k_i^2$, and note that $W = (S_N - 2M_N)/2$. To derive a recurrence for S_N, note that when a node gains one link ($k \to k+1$), its contribution to S changes by $(k+1)^2 - k^2 = 2k+1$. At step $N \to N+1$, the target u always gains degree 1, and in the "all" event each neighbour w of u also gains degree 1. On the other hand, the new node enters with degree $k_v = 1$ or $1 + k_u$. Summing up all contributions, the expected change in S given target u is

$$\mathbb{E}[\Delta S \mid u] = (1-p)\cdot 1 + p(1+k_u)^2 + (2k_u+1) + p\sum_{w\sim u}(2k_w+1), \quad (13)$$

Notice that

$$\sum_u \sum_{w\sim u} k_w = \sum_i k_i^2 = S, \quad (14)$$

as each node w appears as a neighbour of exactly k_w other nodes. Thus, averaging over the uniform choice of u, and using $\sum_u k_u = 2M$, $\sum_u k_u^2 = S$, and (14) gives

$$S_{N+1} = \left(1 + \frac{3p}{N}\right) S_N + 2 + \frac{(4+6p)M_N}{N}. \quad (15)$$

Solving (15) by the same method used for (1) and (10) gives, for $p \neq 0, \frac{1}{2}, \frac{1}{3}$,

$$S = \frac{(6+2p)N}{(1-2p)(1-3p)} + \frac{(4+6p)\mathcal{A}_N}{p(1-2p)} - \frac{4(1+2p)\mathcal{B}_N}{p(1-3p)}, \quad (16)$$

where $\mathcal{A}_N$ and $\mathcal{B}_N$ are as in (11). Recall that $W = (S-2M)/2$, so substituting (2) and (16) yields

$$W = \frac{2(1+2p)}{p}\left[\frac{pN}{(1-2p)(1-3p)} + \frac{\mathcal{A}_N}{1-2p} - \frac{\mathcal{B}_N}{1-3p}\right], \quad (17)$$

Comparing with $3T$ from (11),

$$3T = 6\left[\frac{pN}{(1-2p)(1-3p)} + \frac{\mathcal{A}_N}{1-2p} - \frac{\mathcal{B}_N}{1-3p}\right], \quad (18)$$

we see that W and $3T$ have the same bracketed term, and thus we get

$$\boxed{C = \frac{3p}{1+2p}}, \quad (19)$$

for all $p \in (0,1)$ as $N \to \infty$. In particular, this result is independent of N, which is a hallmark of small-world networks.

FIG. 2. Global clustering coefficient C as a function of p for $N = 10^3$. Error bars represent one standard deviation over 10^3 realisations. The dashed line is the analytical prediction (19).

VI. ASSORTATIVITY

Let X, Y be the random variables representing the degrees of two nodes of an edge chosen uniformly at random from the network. The degree assortativity coefficient is the Pearson correlation coefficient of the degree of pairs of adjacent nodes:

$$r = \frac{\mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])]}{\sigma_X \sigma_Y} = \frac{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}{\mathbb{E}[X^2] - (\mathbb{E}[X])^2}.$$

Note that

- $\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1}{|E|}\sum_{(u,v)\in E} \frac{1}{2}(k_u + k_v)$,
- $\mathbb{E}[X^2] = \mathbb{E}[Y^2] = \frac{1}{|E|}\sum_{(u,v)\in E} \frac{1}{2}(k_u^2 + k_v^2)$,
- $\mathbb{E}[XY] = \frac{1}{|E|}\sum_{(u,v)\in E} k_u k_v$.

Thus we have

$$r = \frac{\frac{1}{|E|}\sum_{(i,j)\in E} k_i k_j - \frac{1}{|E|^2}\left[\sum_{(i,j)\in E}\frac{1}{2}(k_i+k_j)\right]^2}{\frac{1}{|E|}\sum_{(i,j)\in E}\frac{1}{2}(k_i^2+k_j^2) - \frac{1}{|E|^2}\left[\sum_{(i,j)\in E}\frac{1}{2}(k_i+k_j)\right]^2} = \frac{4MR - S^2}{2MQ - S^2}, \quad (20)$$

where $M = |E|$ and $S = \sum_i k_i^2$ are defined as before, $Q = \sum_i k_i^3$, and $R = \sum_{(i,j)\in E} k_i k_j$. Let Q_N and R_N denote the corresponding quantities when the network has N nodes.

We first derive a recurrence for Q_N. Note that when a node i gains one edge ($k_i \to k_i + 1$), its contribution to Q changes by $(k_i+1)^3 - k_i^3 = 3k_i^2 + 3k_i + 1$. At step $N \to N+1$, the target u always gains degree 1, and in the "all" event each neighbour w of u also gains degree 1. On the other hand, the new node enters with degree $k_v = 1$ or $1 + k_u$. Summing up all contributions, the expected change in Q given target u is

$$\mathbb{E}[\Delta Q \mid u] = (1-p)\cdot 1 + p(1+k_u)^3 + (3k_u^2+3k_u+1) + p\sum_{w\sim u}(3k_w^2+3k_w+1). \quad (21)$$

By the same argument as (14), $\sum_u \sum_{w\sim u} k_w^2 = \sum_w k_w^3 = Q$. Thus, averaging over the uniform choice of u, and using $\sum_u k_u = 2M$, $\sum_u k_u^2 = S$, $\sum_u k_u^3 = Q$, and (14) yields

$$Q_{N+1} = \left(1 + \frac{4p}{N}\right) Q_N + 2 + \frac{(3+6p)S_N + (6+8p)M_N}{N}. \quad (22)$$

To derive a recurrence for R_N, note that when node i gains one edge, the product $k_i k_j$ on each incident edge (i,j) increases by k_j. If both endpoints gain degree simultaneously, the product increases by $k_i + k_j + 1$ instead. Write $\pi_i = \sum_{j\sim i} k_j$. We consider the two events separately.

"None" event. Only the target u gains degree 1, and the single new edge (v,u) contributes $k_v \cdot k_u^{\text{new}} = k_u + 1$. The total change is thus

$$\Delta R_{\text{none}} = \pi_u + k_u + 1. \quad (23)$$

"All" event. The target u and all its neighbours gain degree 1, so the set of nodes whose degree increases is $\{u\} \cup N(u)$. Each node i that gains degree 1 increases the product $k_i k_j$ by k_j on every incident edge (i,j), contributing π_i in total, summing over all such nodes yields

$$\pi_u + \sum_{w\sim u} \pi_w. \quad (24)$$

Also note that, for an edge (i,j) where both endpoints gain degree 1, the change in the product is $(k_i+1)(k_j+1) - k_i k_j = k_i + k_j + 1$, but (24) counts only $k_i + k_j$. Thus each such edge requires an additional $+1$ correction. The edges with both endpoints in $\{u\} \cup N(u)$ are the k_u edges (u,w) and the e_u edges among neighbours of u, so we need to add $k_u + e_u$ to the correction. For the new edges, the new node has degree $k_v = 1 + k_u$, so their total contribution to R is

$$(k_u+1)^2 + \sum_{w\sim u}(k_u+1)(k_w+1) = (k_u+1)(2k_u+1+\pi_u).$$

Combining all three parts, we get

$$\Delta R_{\text{all}} = (2+k_u)\pi_u + \sum_{w\sim u} \pi_w + 2k_u^2 + 4k_u + 1 + e_u. \quad (25)$$

Thus the expected change in R given target u is

$$\mathbb{E}[\Delta R \mid u] = (1-p)\,\Delta R_{\text{none}} + p\,\Delta R_{\text{all}}. \quad (26)$$

Note that

$$\sum_u \sum_{w\sim u} \pi_w = \sum_w k_w\,\pi_w = 2R, \quad (27)$$

as each edge $\{w, z\}$ contributes $k_w k_z$ once on each endpoint. Averaging over the uniform choice of u using (27), $\sum_u \pi_u = S$, $\sum_u k_u \pi_u = 2R$, $\sum_u e_u = 3T$, $\sum_u k_u = 2M$, and $\sum_u k_u^2 = S$ yields

$$R_{N+1} = \left(1 + \frac{4p}{N}\right) R_N + 1 + \frac{(1+3p)S_N + (2+6p)M_N + 3pT_N}{N}. \quad (28)$$

Both recurrences (22) and (28) share the homogeneous growth factor $(1 + 4p/N)$, with solution $\mathcal{C}_N = \Gamma(N + 4p)/[\Gamma(1+4p)\,\Gamma(N)]$. Substituting the exact formulas for M_N, S_N, T_N from Sections III–V into the forcing terms and solving by the same method used for (1),(10), and (15) yields, for $p \neq 0, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$,

$$Q_N = \frac{(26+22p)N}{(1-2p)(1-3p)(1-4p)} - \frac{(12+36p+28p^2)\mathcal{A}_N}{2p^2(1-2p)} + \frac{4(3+12p+12p^2)\mathcal{B}_N}{p^2(1-3p)} + \lambda_Q\,\mathcal{C}_N, \quad (29)$$

$$R_N = \frac{3(3+5p)N}{(1-2p)(1-3p)(1-4p)} - \frac{(4+16p+18p^2)\mathcal{A}_N}{2p^2(1-2p)} + \frac{2(2+10p+15p^2)\mathcal{B}_N}{p^2(1-3p)} + \lambda_R\,\mathcal{C}_N, \quad (30)$$

where $\mathcal{A}_N$ and $\mathcal{B}_N$ are as in (11), and λ_Q, λ_R are constants determined by the initial conditions $Q_1 = R_1 = 0$.

Since $\mathcal{C}_N \sim N^{4p}$, for $p < 1/4$, the homogeneous terms $\mathcal{A}_N$, $\mathcal{B}_N$, $\mathcal{C}_N$ are all sublinear and all five quantities M, S, T, Q, R grow linearly. Let

$$\alpha_Q = \frac{26+22p}{(1-2p)(1-3p)(1-4p)}, \quad (31)$$

$$\alpha_R = \frac{3(3+5p)}{(1-2p)(1-3p)(1-4p)} \quad (32)$$

be the linear term coefficients of Q_N and R_N respectively. From (1) and (16) we can also see that $\alpha_M = 1/(1-2p)$ and $\alpha_S = (6+2p)/[(1-2p)(1-3p)]$. For $p < 1/4$, we have

$$4\alpha_M\alpha_R - \alpha_S^2 = \frac{8p(9-2p)(1-p)}{(1-2p)^2(1-3p)^2(1-4p)},$$

$$2\alpha_M\alpha_Q - \alpha_S^2 = \frac{8(1-p)(1+2p)(2-p)}{(1-2p)^2(1-3p)^2(1-4p)}. \quad (33)$$

It now follows that as $N \to \infty$,

$$\boxed{r \sim \frac{p(9-2p)}{(1+2p)(2-p)}} \quad (34)$$

for $p < 1/4$.

FIG. 3. Assortativity coefficient r as a function of p for $N = 10^3$. The dashed line is the analytical prediction (34).

VII. AVERAGE SHORTEST-PATH DISTANCE

Let $D = \sum_{i<j} d(i,j)$ denote the expected sum of all pairwise shortest-path distances, and write D_N for this sum when the network has N nodes. The average shortest-path distance is then

$$\ell = \frac{2D_N}{N(N-1)}. \quad (35)$$

At step $N \to N+1$, a target u is chosen uniformly from the N existing nodes. Note that adding v does not shorten any existing pairwise distance, as any path through v can be rerouted through u or its neighbours without increasing length. Thus

$$D_{N+1} = D_N + \sum_j d(v,j). \quad (36)$$

Write $\sigma_u = \sum_{j\neq u} d(u,j)$. We now consider the two events separately.

"None" event. The new node v is connected only to the target u. Thus, $d(v,u) = 1$ and $d(v,j) = 1 + d(u,j)$ for every $j \neq u$, which yields

$$\sum_j d(v,j) = N + \sigma_u. \quad (37)$$

"All" event. The new node v is adjacent to the target u and all its neighbours. But then $N(u) = N(v)$, so any path from u to j can be rerouted through v by simply replacing the starting point u with v. Thus, $d(v,j) = d(u,j)$ for all $j \neq u$, and so

$$\sum_j d(v,j) = 1 + \sigma_u. \quad (38)$$

Averaging over the two events and the uniform choice of u, and noting that $\frac{1}{N}\sum_u \sigma_u = 2D_N/N$, we have

$$\sum_j d(v,j) = (1-p)(N + \mathbb{E}_u[\sigma_u]) + p(1 + \mathbb{E}_u[\sigma_u]) = (1-p)N + p + \frac{2D_N}{N}, \quad (39)$$

and so

$$D_{N+1} = \left(1 + \frac{2}{N}\right) D_N + (1-p)N + p. \quad (40)$$

The homogeneous solution is $\mathcal{D}_N = N(N+1)/2$, satisfying $\mathcal{D}_{N+1}/\mathcal{D}_N = (N+2)/N$. Set $D_N = \gamma_N \mathcal{D}_N$. Then (40) is reduced to

$$\gamma_{N+1} - \gamma_N = \frac{2[(1-p)N+p]}{(N+1)(N+2)} = \frac{2(2p-1)}{N+1} + \frac{2(2-3p)}{N+2}.$$

It follows from $\gamma_1 = 0$ that

$$\gamma_N = \sum_{k=1}^{N-1} \frac{2(2p-1)}{k+1} + \frac{2(2-3p)}{k+2} \tag{41}$$

$$= 2(2p-1)(H_N - 1) + 2(2-3p)\left(H_{N+1} - \frac{3}{2}\right), \tag{42}$$

where $H_N = \sum_{j=1}^{N} 1/j$ is the N-th harmonic number. Since $H_N \sim \ln N$ as $N \to \infty$, we have $\gamma_N \sim 2(1-p)\ln N$, and so

$$D_N \sim (1-p)\, N(N+1) \ln N. \tag{43}$$

Substituting into (35) now yields

$$\boxed{\ell \sim 2(1-p)\ln N}, \tag{44}$$

for all $p \in [0,1)$. At $p = 1$ the network is a complete graph with $\ell = 1$. The distance scales logarithmically for every with no phase transition, and thus confirms the small-world property that we established with the $O(1)$ clustering coefficient from Section V.

FIG. 4. Average shortest-path distance ℓ as a function of N on a log-linear scale for several values of p. Dashed lines show the analytical prediction $2(1-p)\ln N$.

VIII. CONCLUSION

We have studied the all-or-none network growth model and derived exact or asymptotic analytical expressions for five structural properties, all controlled by the single parameter p:

(i) The expected number of edges transitions from linear ($\sim N/(1-2p)$ for $p < 1/2$) through $N \ln N$ at $p = 1/2$ to superlinear ($\sim N^{2p}$) for $p > 1/2$. (ii) The degree distribution follows a power law $P_k \sim k^{-1/p}$ with a continuously tunable exponent, matching the independent copying model at the mean-field level. (iii) The global clustering coefficient converges to $C = 3p/(1+2p)$, a remarkably simple N-independent expression valid for all p. (iv) In the sparse regime ($p < 1/3$), the assortativity coefficient is $r = p(9-2p)/[(1+2p)(2-p)]$, derived by moment closure, increasing from $r = 0$ at $p = 0$ to $r \to 1$ as $p \to 1/3^-$. (v) An exact topological recurrence for the sum of pairwise distances shows that the average shortest-path distance scales as $\ell \sim 2(1-p)\ln N$ for all $p < 1$, with no phase transition.

The network is a *small world* for every $p \in (0,1)$: logarithmic distances coexist with $O(1)$ clustering. The parameter p smoothly interpolates between a random recursive tree ($p = 0$) and the complete graph ($p = 1$), passing through regimes that reproduce key features of real-world networks including scale-free degree distributions, tuneable assortativity, and the small-world property. Future directions include extending the assortativity formula beyond $p = 1/3$, analysing the model on directed or weighted networks, and studying its spectral properties.

ACKNOWLEDGMENTS

The author thanks Professor R. Lambiotte for helpful discussions and the problem formulation.

[1] A.-L. Barabási and R. Albert, Emergence of scaling in random networks, Science **286**, 509 (1999).
[2] M. E. J. Newman, The structure and function of complex networks, SIAM Review **45**, 167 (2003).
[3] J. Kleinberg, R. Kumar, P. Raghavan, S. Rajagopalan, and A. Tomkins, The web as a graph: Measurements, models, and methods, in *International Computing and Combinatorics Conference* (Springer, 1999) pp. 1–17.